STAT 418 -- FINAL PROJECT

Airline Merger Damages Estimator

Estimating Consumer Harm from the AA / US Airways Merger using Causal Machine Learning

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Project Overview & Motivation

What This Project Does

- Replicates antitrust damages estimation work done by economic consulting firms in litigation.
- Quantifies consumer harm from the 2013 AA / US Airways merger using the but-for methodology standard in expert witness testimony.

Core Research Questions

- Did the merger raise ticket prices on routes where AA and US Airways previously competed?
- What is the total dollar value of consumer overcharges?
- Which routes and markets were harmed most?

Target Users

- Antitrust economists and policy researchers
- Airline market analysts evaluating domestic consolidation

Dec 2013

Merger closed

1,223

Routes where AA + US competed

622M

Affected passengers (2014-2017)

Data Collection Approach & Progress

Primary Data: DOT DB1B (BTS)

- Source: BTS Origin & Destination Survey
- Method: Programmatic download, no API key required
- 32 quarters (2010-2017), ~5M ticket records per quarter
- Final panel: 122,787 route-quarter observations

```
# Automated bulk download loop
url = f"transtats.bts.gov/PREZIP/"
    f"DB1BMarket_{year}_{quarter}.zip"
```

Control Variables: FRED API

- Jet fuel prices: Primary cost confounder
- Airline CPI: Macro price trend
- Free FRED API, single authenticated call per series

32	4 GB	2
Quarters downloaded	Raw data on disk	FRED series pulled

Sample DB1B Record

Route	Carrier	Fare ($)	Pax
LAX-JFK	AA	452	41
PHL-CLT	US	189	28

Model Development & Evaluation

STANDARD	PRIMARY	HETEROGENEITY
Two-Way FE	**Double ML (LightGBM)**	**Causal Forest**
Route + time fixed effects. Econometrics baseline.	Partials out nonlinear cost confounders.	Route-level treatment effect variation.
ATE: -2.54% (Spurious)	**ATE: +2.57% (Causal)**	**ATE: +4.48% (ITE mean)**

Key Finding — Sign Flip: The 2014-2015 fuel crash confounded linear estimators, producing a spurious negative ATE. Double ML correctly partials out nonlinear controls via LightGBM, recovering +2.57%.

Damages Calculation

- (Actual fare - Counterfactual fare) x Passengers = Consumer Harm
- Counterfactual fare = Actual fare / (1 + 0.0257)
- 95% CI on ATE: [+1.13%, +5.44%] — statistically significant

$3.97B
Total Consumer Harm

$6.73
Avg Overcharge / Ticket

622M
Affected Passengers

Solution Architecture

DOT DB1B API
+ FRED API

Data Pipeline
(Python)

Model Training
(Double ML + CF)

DATA & MODELING

Model Artifacts (joblib .pkl)

Flask API
(Cloud Run / 8080)

Streamlit App
(Cloud Run / 8501)

End User
(Browser)

DEPLOYMENT

Stack

- Python 3.11 | EconML | LightGBM | joblib
- Flask | Streamlit | Plotly
- Docker | Google Cloud Run

App: https://merger-app-1010508609944.us-central1.run.app

API: https://merger-api-1010508609944.us-central1.run.app/health

Application Features & User Interface

Executive Summary

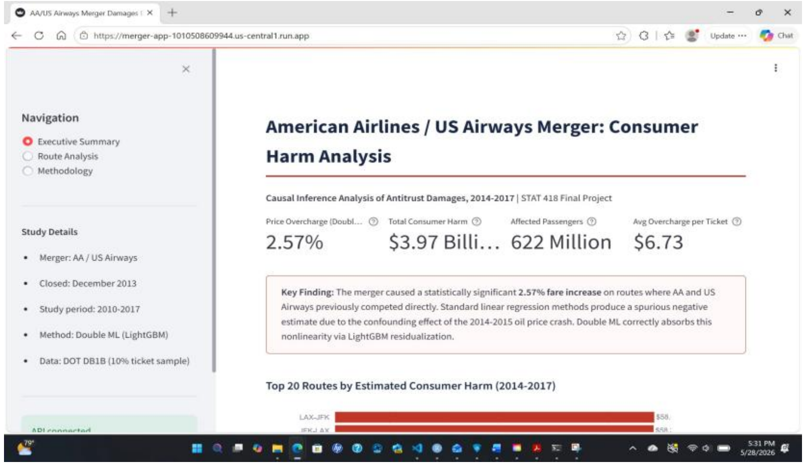
- 4 headline metrics: 2.57%, $3.97B, 622M, $6.73
- Top 20 routes bar chart
- Model comparison table — sign flip explanation
- Full ranked table of all 1,223 treated routes

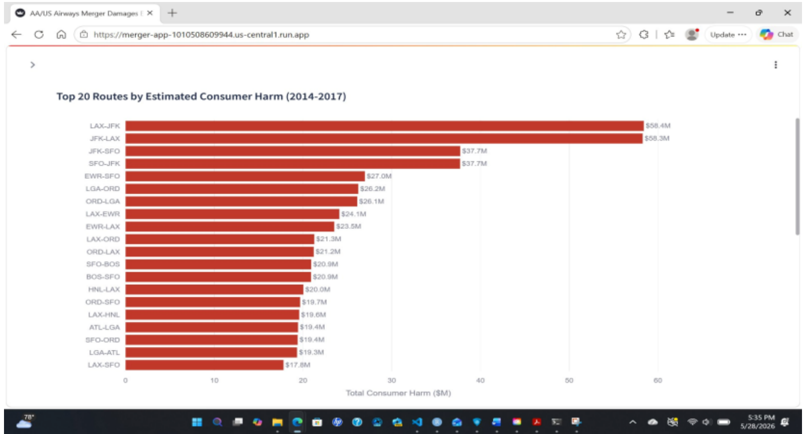
Route Analysis

- Dropdown: 200 treated routes, default LAX-JFK
- Actual vs. counterfactual fare metrics
- Fare comparison bar chart (red vs. navy)
- Fare history 2010-2017 with 95% CI band

Methodology

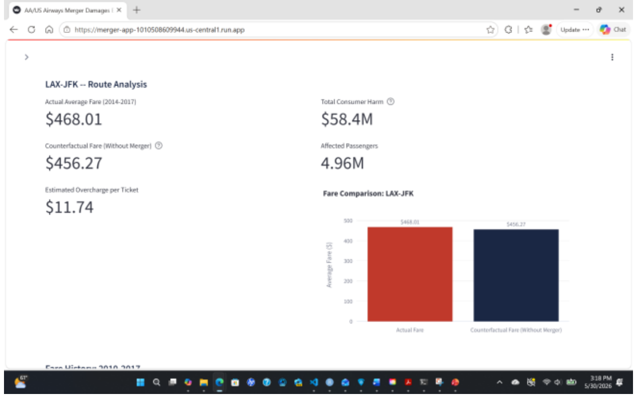
- Causal identification strategy explained
- Why Double ML? — fuel price confound
- Three-step Double ML procedure
- Damages calculation formula + data sources





Application Example Walkthrough

Route Analysis — LAX-JFK

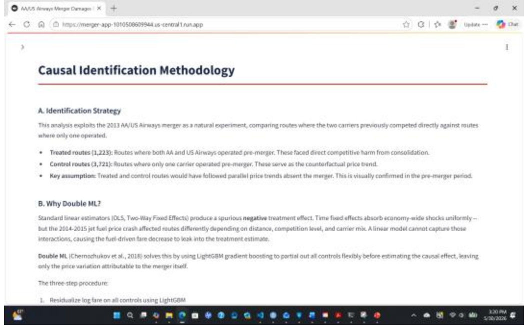


Select any treated route — actual fare, counterfactual, overcharge, total harm

Fare History with 95% CI Band

Pre/post-merger fare trends with Double ML confidence interval band

Methodology Page



Live API Endpoint Example

```
POST /predict
{"route": "LAX-JFK"}
-> overcharge: $11.74, harm: $58.4M
```

Flask API deployed on Google Cloud Run — 4 endpoints, full CORS

Challenges, Lessons Learned & AI Workflow

Technical Challenges

Sign flip — the central finding

OLS/TWFE produced a spurious negative ATE due to the 2014-2015 fuel price crash coinciding with the post-merger period. Resolved by switching to Double ML.

EconML API incompatibility

ate_inference().summary_frame() unavailable in newer EconML versions. Fixed by using coef__interval() instead.

Cloud Run build context

Dockerfile COPY paths assumed project root but --source deployment uses subfolder as context. Fixed by copying artifacts locally before deploy.

Dual processes on port 8080

Stale Flask process blocked new deployment. Diagnosed via netstat, killed both PIDs, restarted clean.

AI-Assisted Development Workflow

Claude (chat): Refining methodology, specification, debugging

Claude Code: Notebooks, Flask API, Streamlit app, Dockerfiles

Key Lessons

- CLAUDE_CONTEXT.md as a living spec dramatically improved Claude Code output — specific constraints prevented recurring errors
- Separating architecture design (Claude chat) from code generation (Claude Code) produced better results than doing both in one tool
- AI code still required human review — EconML API changes, Windows encoding, and Docker path issues all needed manual diagnosis